Clinical Immunology - Structural Correspondence with the 5-Stage Model

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Clinical Immunology

Survey of structural correspondence with the 5-stage model (Field · Wave · Edge · Vortex · Bundle)

Survey Overview

- **Scope**: 10 major theories in clinical immunology
- **Research question**: Do clinical-immunology theories correspond structurally to the 5-stage model?
- **Results**: Strong correspondence in 8 cases, partial correspondence in 2 cases

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Structural Correspondence Diagram

Overview of the 5-Stage Model

Stage	Definition
Field (ba)	Undifferentiated state. Initial condition in which neither direction nor structure has yet been determined
Wave (nami)	Exploration stage in which multiple directions diverge and compete
Edge (en)	Tension state in which opposing elements coexist without converging on either side. The place where things meet at a boundary, influence each other, and relationships are formed
Vortex (uzu)	Stage in which new coherence (order) spontaneously emerges from tension
Bundle (taba)	Stage in which form is fixed and stabilized as a reusable structure

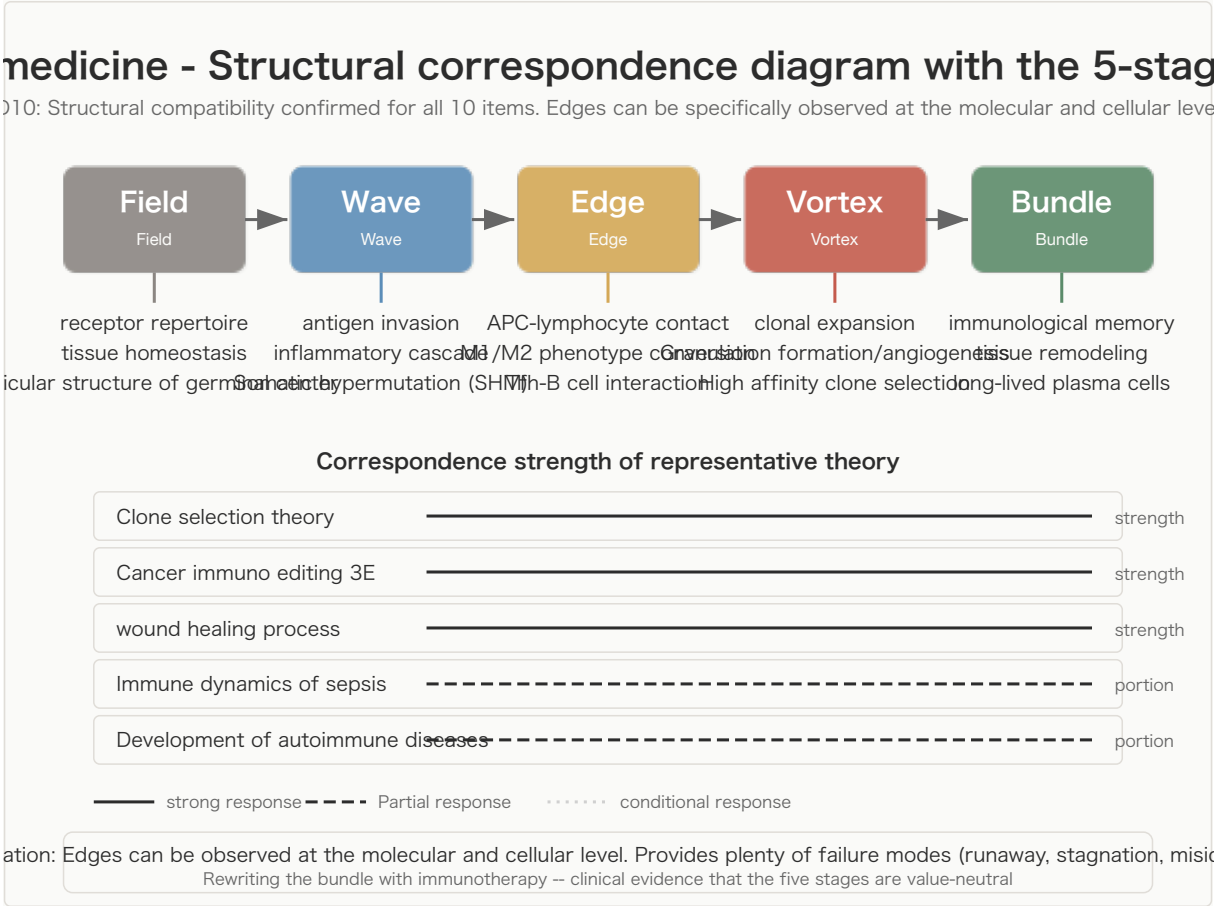

Figure 1: Structural correspondence diagram

Overview of Structural Correspondences

Figure 2: Domain — Theory x 5-Stage Correspondence Matrix

#	Theory / Process	Proponents	Corresponding Stages	Judgment
1	Clonal selection theory	Burnet (1957), Tonegawa (1983)	Field → Wave → Edge → Vortex → Bundle	Strong
2	Cancer immunoediting 3E	Dunn, Schreiber, Old (2002–2004)	Field → Wave → Edge → Vortex → Bundle	Strong
3	Active resolution of inflammation	Serhan (2000s–)	Field → Wave → Edge → Vortex → Bundle	Strong
4	Wound healing process	Guo & DiPietro (2010), Landén et al. (2016)	Field → Wave → Edge → Vortex → Bundle	Strong
5	Immune dynamics in sepsis	Bone (1996), Hotchkiss et al. (2013)	Field → Wave → Edge → Vortex → Bundle	Partial
6	Fracture repair process	Einhorn & Gerstenfeld (2015)	Field → Wave → Edge → Vortex → Bundle	Strong
7	Allergen immunotherapy	Akdis & Akdis (2014)	Field → Wave → Edge → Vortex → Bundle	Strong

#	Theory / Process	Proponents	Corresponding Stages	Judgment
8	Transplant immunity and tolerance induction	Medawar (1944), Kawai et al. (2008)	Field → Wave → Edge → Vortex → Bundle	Strong
9	Pathogenesis of autoimmune disease	Rose & Bona (1993), Rosenblum et al. (2015)	Field → Wave → Edge → Vortex → Bundle	Partial
10	Germinal center reaction and affinity maturation	MacLennan (1994), Victora & Nussenzweig (2012)	Field → Wave → Edge → Vortex → Bundle	Strong

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Key Entry 1: Adaptive Immune Response by Clonal Selection Theory (Burnet, Tonegawa)

- Clonal selection theory is the central principle of adaptive immunity. Through genetic recombination (V(D)J recombination), lymphocytes with up to 10^{11} diverse antigen receptors are generated within the body; only the clones that match an antigen are selectively expanded to form effector and memory cells. Burnet (1957) proposed the principle, and Tonegawa (1983) elucidated the molecular mechanism.
- **As fact**: V(D)J recombination is a mechanism that deliberately cleaves and rejoins lymphocyte genes to generate enormous receptor diversity. When an antigen enters, clones bearing receptors that match it are activated and expand/differentiate to form an effector response. After the response, some clones are maintained long-term as memory cells and display a faster, stronger response upon secondary infection (Murphy & Weaver, 2016).
- **As reading**: Here, we read a mechanism that might be called "institutionalized error production." The immune system deliberately cleaves DNA and mass-produces random combinations in order to prepare in advance a "latent space capable of responding to anything that arrives." The level of similarity is mechanism, with particular attention to the causal sequence: random pre-generation of diversity → selection by the environment → amplification → fixation.
- **As interpretation**: The receptor repertoire generated by V(D)J recombination — a latent space that has not yet responded to anything — corresponds to the **Field**. The process by which specific clones become reactive upon antigen entry corresponds to the **Wave**. The process in which antigen-presenting cells (dendritic cells and macrophages) and T/B cells physically contact each other via MHC-peptide complexes and exchange signals corresponds to the **Edge**. The proliferation and differentiation of clones as a result of that contact corresponds to the **Vortex**. The state of long-term maintenance as memory cells corresponds to the **Bundle**.
- What makes this correspondence noteworthy is that the immune system institutionalizes, as "planned error production," the deliberately harmful operation of cleaving DNA. Diversity is not the product of chance but of institutional production, and only what proves adaptive to the environment is selected.

Key Entry 2: Cancer Immunoediting 3E (Dunn, Schreiber, Old)

- Cancer immunoediting is a model describing the long-term co-evolution of the immune system and tumors. Dunn, Old, and Schreiber (2004) proposed three phases: Elimination, in which the immune system eliminates tumor cells; Equilibrium, in which tumor cells change under immune pressure while coexisting; and Escape, in which clones that have evaded immunity grow out.
- **As fact**: The immune system has the ability to recognize and eliminate tumor cells, but simultaneously, under immune pressure, tumor cells are selected and clones that evade immune recognition emerge. The Equilibrium phase can last years, persisting clinically as latent cancer. The clinical success of immune checkpoint inhibitor therapy (e.g., PD-1/PD-L1 inhibitors) supports the validity of this model (Sharma & Allison, 2015).

- **As reading**: Here, we read a structure in which two different agents (the immune system and the tumor) exert mutual selective pressure over a long period, and a coexistent state in which neither gains a decisive upper hand persists. The level of similarity is process, with attention to the temporal unfolding: "long-term coexistence of opposing forces → disruption of equilibrium through escape by one side."
- **As interpretation**: The coexistence of intratumoral genetic diversity and immune surveillance corresponds to the **Field**. The process in which the immune system recognizes neoantigens (tumor-specific mutant proteins) and differences among tumor cells become manifest corresponds to the **Wave**. The interaction between T cells and tumor cells, the relationship between NK cells and MHC molecules, and the antagonism between suppression and activation mediated by PD-1/PD-L1 correspond to the **Edge**. The Equilibrium phase in particular is notable as a "prolonged Edge" — settled neither into elimination nor into escape. The selective expansion of immune-escape clones corresponds to the **Vortex**. Fixation as a progressed tumor, or elimination/remission through immunotherapy, corresponds to the **Bundle**.
- The important insight here is that the Vortex is not necessarily "constructive." The emergence of immune-escape clones is a creation harmful to the organism, suggesting that what the 5 stages describe is not "good creation" but "creation as such" — that is, the 5 stages are value-neutral.

Key Entry 3: Wound Healing Process (Guo & DiPietro, Landén et al.)

- Wound healing is a sequential process through which tissue is repaired via hemostasis, inflammation, cellular proliferation, and remodeling. Transitions between stages are regulated at the molecular level; in particular, the phenotypic switching of macrophages is known to drive transitions between stages.
- **As fact**: Upon injury, platelet-mediated hemostasis and fibrin clot formation occur first, releasing growth factors such as PDGF and TGF- . Neutrophils and macrophages then infiltrate and the inflammatory response unfolds. Macrophages switch phenotype from pro-inflammatory (M1) to pro-repair (M2), and this switch promotes fibroblast activation, angiogenesis, and re-epithelialization (Landén et al., 2016). In the final stage, remodeling progresses through collagen reorganization and cross-linking, stabilizing tissue structure. Note that the M1/M2 dichotomy represents a model for the two ends of a continuous spectrum, and actual macrophage phenotypes can adopt more diverse mixed states (Murray et al., 2014).
- **As reading**: Here, we read a structure in which the same cell type (macrophage), in response to its microenvironment, switches between diametrically opposed roles — pro-inflammatory and pro-repair — and this switching drives the transition to the next stage. The level of similarity is process, with attention to the "temporal ordering of stages" and the causal structure in which "the role-switching of a single agent drives stage transitions."
- **As interpretation**: The hemostatic clot forming in the wound bed (a reservoir of fibrin and growth factors) corresponds to the **Field**. The time-staggered infiltration of inflammatory cells — from neutrophils to macrophages — corresponds to the **Wave**. The multiplex interactions among macrophages (centered on the M1-to-M2 phenotypic switch), fibroblasts, and endothelial cells correspond to the **Edge**. The amplification of granulation tissue formation, angiogenesis, and re-epithelialization through positive feedback corresponds to the **Vortex**. Structural stabilization through collagen reorganization, cross-linking, and wound contraction corresponds to the **Bundle**.
- The most important insight that wound healing offers is that the Edge is not merely "relationship" but a "site of qualitative transformation." The fact that the same macrophage switches from a destructive role to a healing role depending on its environment concretely demonstrates that the essence of the Edge lies in the dynamics of relationship. Moreover, the clinical picture in which failure of the M1-to-M2 switch leads to chronic wounds (e.g., diabetic ulcers) is a concrete example of the failure mode in which a failed transition at the Edge blocks progress toward the Vortex.

Cross-Cutting Patterns

- The most striking cross-cutting feature of clinical medicine is the "observability of the Edge"
- Second, a "systematicity of failure modes" emerges

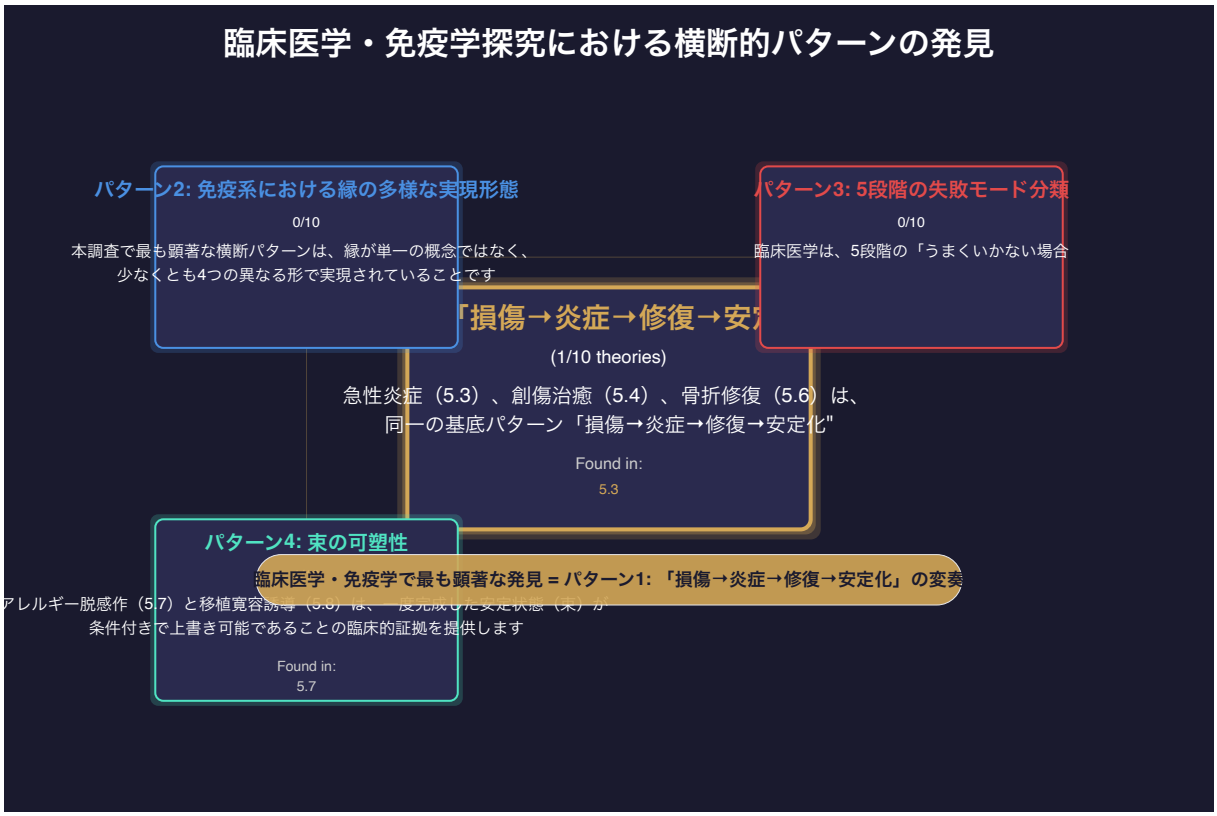

Figure 3: Domain — Cross-Cutting Patterns Diagram

- Third, a "variation on tissue repair" — a shared pattern of "injury → inflammation → repair → stabilization" common to wound healing and fracture repair — is confirmed
- Fourth, "plasticity of the Bundle" is confirmed across multiple processes

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Open Questions

- In the immune system, "Edge" is used at different levels: physical cell-to-cell contact (antigen presentation), the self/non-self recognition boundary, and the branch point of therapeutic intervention. Whether it is appropriate to treat all of these as the same "Edge," or whether sub-classification is needed, requires further clarification.
- The SIRS/CARS model of sepsis has been criticized as "overly simplified"; it is now known that SIRS and CARS actually progress simultaneously from early on. When mapping this model onto the 5 stages, how to describe a concurrent rather than sequential dynamic is a remaining challenge.
- Applying the 5 stages to pathological processes (cancer immune escape, autoimmunity, sepsis) implies extending the model's descriptive scope from "normal creation" to "structural change in general." The validity and limits of this extension require cross-referencing with findings from other domains.
- Clonal selection theory and the germinal center reaction can be read as a nested structure at macro and micro scales, but whether this "repetition of the same structure at different scales" is a property specific to the 5 stages or a reflection of a more general self-organizing phenomenon requires verification.

Conclusion

- This survey confirmed that clinical medicine and immunology, as a whole, form a domain with strong structural correspondence to the 5-stage model
- Three particularly important findings stand out

- At the same time, immunological findings are concentrated at the molecular and cellular scale; whether the 5-stage correspondence is particularly suited to this scale, or whether equivalent correspondence can be obtained at other scales as well, requires careful examination